Matteo Spezialetti, Ph.D.

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Work Experience

- 2025 – present
  Position: Research Fellow
  Organization: Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila
  Project: Optimization, deep learning and signal processing techniques for E-Health
  P.I.: Prof. Filippo Mignosi
- 2020 – 2025
  Position: Researcher (RTD-A)
  Organization: Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila
  Project: Optimization of computational resources in TPS (Treatment Planning System) for hadron therapy
  P.I.: Prof. Filippo Mignosi
- 2019 – 2020
  Position: Research Fellow
  Organization: Department of Electrical Engineering and Information Technology, University "Federico II" of Naples
  Project: Automatic evaluation of the experience during multi-agent human-machine and human-robot interaction
  P.I.: Prof. Silvia Rossi
- 2018 – 2019
  Position: Research Fellow
  Organization: Department of Psychology, University "La Sapienza" of Rome
  Project: The body in the brain at rest: MEG studies
  P.I.: Prof. Viviana Betti
- 2013
  Position: Term-contract worker
  Organization: Department of Life, Health and Environmental Sciences, University of L'Aquila
  Project: Design and implementation of a communication system for people with lacking sensory/motor skills, based on the analysis of EEG signals
  P.I.: Prof. Giuseppe Placidi

Bibliometric Indicators

- updated at 13/10/2026

	Scopus	Web of Science
Journal articles	23	23
Conference papers	33	27
Total documents	**56**	**50**
Total citations	**1235**	**901**
h-index	**19**	**18**

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Teaching

- 2021/2022 – present • Compilers: Fo151, 6 CFU, Computer Science, DISIM, UNIVAQ
- 2025 – present • Co-Advisor of a student of the PhD Program in Experimental Medicine, DISCAB, UNIVAQ
- 2024 • Mälardalen University (ERASMUS+ teaching agreement), 10 hours:
  - ICT and ML for signal and image processing (2 hours)
  - Deep Learning for Dose Refinement in Proton Therapy (4 hours)
  - Optimization aspects in Proton Therapy (4 hours)
- 2021/2022 – 2023/2024 • Computer Science Skill: Bo487, 3 CFU, Biotechnologies, DISCAB, UNIVAQ
- 2024 – present • Co-Advisor of two students of the PhD Program in Information and Communication Technology, DISIM, UNIVAQ
- 2020/2021 • Introduction to Computability and Complexity: Fo150, 6 CFU, Computer Science, DISIM, UNIVAQ
- 2020/2021 • Information Retrieval: DTo211, 3 CFU, Computer Science, DISIM, UNIVAQ
- 2021/2022 • Compilers: Fo151, 6 CFU, Computer Science, DISIM, UNIVAQ, (Owner Teacher: Prof. Paola Inverardi)
- 2019/2020 • Theory of Languages: DTo269, 6 CFU, Computer Science, DISIM, UNIVAQ
- 2019/2020 • Compilers Laboratory: DTo275, 3 CFU, Computer Science, DISIM, UNIVAQ
- 2020 – 2025 • Ph.D. Board Member: Health and Environmental Sciences, MESVA, UNIVAQ
- 2020 – present • Tutor or advisor of students of bachelor and master degrees in Computer Science, Information Engineering and Mathematics

Education

- 2018 • **Degree:** Ph.D. in Health and Environmental Sciences (Molecular and Ultrastructural Imaging
  Organization: Department of Life, Health and Environmental Sciences, University of L'Aquila
  Thesis title: Emotion driven brain computer interfaces: alternative pathways for people in minimally conscious state
  Advisor: Prof. Giuseppe Placidi
- 2016 • **Position:** Visiting Student
  Organization: Faculty of Engineering, University of Porto
  Tutor: Prof. João Manuel RS Tavares
- 2014 • **Degree:** Master's Degree in Computer Science (110/110 with honors)
  Organization: Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila
  Tutor: Prof. Giuseppe Placidi
- 2012 • **Degree:** Bachelor's Degree in Computer Science (107/110)
  Organization: Faculty of Science, University of L'Aquila
  Tutor: Prof. Giuseppe Placidi

Scientific Publications

Journal Articles

1 D. Fossemò, F. Mignosi, G. Placidi, L. Raggioli, M. Spezialetti, and F. A. D'ASARO, "Explaining neural networks in preference learning: A post hoc inductive logic programming approach," *Theory and Practice of Logic Programming*, pp. 1–49, 2026.

2 O. Maddaluno, S. Della Penna, A. Pizzuti, et al., "Encoding manual dexterity through modulation of intrinsic α band connectivity," *Journal of Neuroscience*, vol. 44, no. 20, 2024.

3 D. Sili, C. De Giorgi, A. Pizzuti, M. Spezialetti, F. de Pasquale, and V. Betti, "The spatio-temporal architecture of everyday manual behavior," *Scientific Reports*, vol. 13, no. 1, p. 9451, 2023.

4 R. Ornello, M. Spezialetti, V. Caponnetto, et al., "Different effects of air microembolism through patent foramen ovale in patients with migraine: A quantitative electroencephalogram case series," *Frontiers in Neurology*, vol. 13, 2022.

5 E. Theodoridou, L. Cinque, F. Mignosi, et al., "Hand tracking and gesture recognition by multiple contactless sensors: A survey," *IEEE Transactions on Human-Machine Systems*, 2022.

6 R. Aragona, F. Marzi, F. Mignosi, and M. Spezialetti, "Entropy and compression: A simple proof of an inequality of khinchin-ornstein-shields," *Problems of Information Transmission*, vol. 56, pp. 13–22, 2020.

7 M. Rescigno, M. Spezialetti, and S. Rossi, "Personalized models for facial emotion recognition through transfer learning," *Multimedia Tools and Applications*, vol. 79, pp. 35 811–35 828, 2020.

8 M. Spezialetti, G. Placidi, and S. Rossi, "Emotion recognition for human-robot interaction: Recent advances and future perspectives," *Frontiers in Robotics and AI*, p. 145, 2020.

9 G. Placidi, M. Polsinelli, M. Spezialetti, L. Cinque, P. Di Giamberardino, and D. Iacoviello, "Self-induced emotions as alternative paradigm for driving brain–computer interfaces," *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, vol. 7, no. 5-6, pp. 512–519, 2019.

10 G. Placidi, L. Cinque, M. Polsinelli, and M. Spezialetti, "Measurements by a leap-based virtual glove for the hand rehabilitation," *Sensors*, vol. 18, no. 3, p. 834, 2018.

11 M. Spezialetti, L. Cinque, J. M. R. Tavares, and G. Placidi, "Towards eeg-based bci driven by emotions for addressing bci-illiteracy: A meta-analytic review," *Behaviour & Information Technology*, vol. 37, no. 8, pp. 855–871, 2018.

12 M. Carrieri, A. Petracca, S. Lancia, et al., "Prefrontal cortex activation upon a demanding virtual hand-controlled task: A new frontier for neuroergonomics," *Frontiers in human neuroscience*, vol. 10, p. 53, 2016.

13 S. B. Moro, M. Carrieri, D. Avola, et al., "A novel semi-immersive virtual reality visuomotor task activates ventrolateral prefrontal cortex: A functional near-infrared spectroscopy study," *Journal of neural engineering*, vol. 13, no. 3, p. 036 002, 2016.

14 G. Placidi, A. Petracca, M. Spezialetti, and D. Iacoviello, "A modular framework for eeg web based binary brain computer interfaces to recover communication abilities in impaired people," *Journal of medical systems*, vol. 40, pp. 1–14, 2016.

15 D. Iacoviello, A. Petracca, M. Spezialetti, and G. Placidi, "A classification algorithm for electroencephalography signals by self-induced emotional stimuli," *IEEE transactions on cybernetics*, vol. 46, no. 12, pp. 3171–3180, 2015.

16 D. Iacoviello, A. Petracca, M. Spezialetti, and G. Placidi, "A real-time classification algorithm for eeg-based bci driven by self-induced emotions," *Computer methods and programs in biomedicine*, vol. 122, no. 3, pp. 293–303, 2015.

17 F. Pistoia, A. Carolei, D. Iacoviello, et al., "Eeg-detected olfactory imagery to reveal covert consciousness in minimally conscious state," *Brain injury*, vol. 29, no. 13-14, pp. 1729–1735, 2015.

18 G. Placidi, D. Avola, A. Petracca, F. Sgallari, and M. Spezialetti, "Basis for the implementation of an eeg-based single-trial binary brain computer interface through the disgust produced by remembering unpleasant odors," *Neurocomputing*, vol. 160, pp. 308–318, 2015.

19 M. Ferrari, S. Bisconti, M. Spezialetti, et al., "Prefrontal cortex activated bilaterally by a tilt board balance task: A functional near-infrared spectroscopy study in a semi-immersive virtual reality environment," *Brain topography*, vol. 27, pp. 353–365, 2014.

20 S. B. Moro, S. Bisconti, M. Muthalib, et al., "A semi-immersive virtual reality incremental swing balance task activates prefrontal cortex: A functional near-infrared spectroscopy study," *Neuroimage*, vol. 85, pp. 451–460, 2014.

21 G. Placidi, D. Avola, M. Ferrari, et al., "A low-cost real time virtual system for postural stability assessment at home," *Computer methods and programs in biomedicine*, vol. 117, no. 2, pp. 322–333, 2014.

22 D. Avola, M. Spezialetti, and G. Placidi, "Design of an efficient framework for fast prototyping of customized human–computer interfaces and virtual environments for rehabilitation," *Computer Methods and Programs in Biomedicine*, vol. 110, no. 3, pp. 490–502, 2013.

Conference Proceedings

1 J. Jawaid, M. Spezialetti, G. Libardi, G. Stilo, and F. Mignosi, "Optimization of deep learning model for denoising dose profiles in proton therapy," in *4th International Conference on Frontiers of Artificial Intelligence, Ethics, and Multidisciplinary Applications (FAIEMA 2026) - ACCEPTED*, 2026.

2 M. Spezialetti, A. D'Angelo, F. Ciccarelli, G. Costanzo, D. Fossemò, and F. Mignosi, "Web&graph 2026 workshop report: Workshop on web & graphs, responsible intelligence, and social media," in *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, 2026, pp. 39–43.

3 M. Spezialetti, A. D'Angelo, F. Ciccarelli, G. Costanzo, D. Fossemò, and F. Mignosi, "Web&graph 2026: Workshop on web & graphs, responsible intelligence, and social media," in *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, 2026, pp. 1390–1391.

4 C. Epifanio, L. Forlizzi, F. Marzi, F. Mignosi, G. Placidi, and M. Spezialetti, "On the k-hamming and k-edit distances," in *ICTCS 2023-Italian Conference on Theoretical Computer Science 2023 Proceedings of the 24th Italian Conference on Theoretical Computer*, CEUR, 2023, pp. 143–156.

5 M. Spezialetti, R. G. De Lorenzo, G. L. Gravina, et al., "Spread-out bragg peak in treatment planning system by mixed integer linear programming: A proof of concept," in *2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS)*, IEEE, 2023, pp. 548–554.

6 D. Fossemò, F. Mignosi, L. Raggioli, M. Spezialetti, and F. A. D'Asaro, "Using inductive logic programming to globally approximate neural networks for preference learning: Challenges and preliminary results," in *CEUR WORKSHOP PROCEEDINGS*, 2022, pp. 67–83.

7 D. Lozzi, F. Mignosi, G. Placidi, M. Polsinelli, and M. Spezialetti, "A 4d lstm network for emotion recognition from the cross-correlation of the power spectral density of eeg signals," in *The International Workshop on Affective Computing and Emotion Recognition (ACER-EMORE2022)*, IEEE/WIC/ACM, 2022.

8 G. Placidi, A. Di Matteo, F. Mignosi, M. Polsinelli, and M. Spezialetti, "Compact, accurate and low-cost hand tracking system based on leap motion controllers and raspberry pi.," in *ICPRAM*, 2022, pp. 652–659.

9 M. Spezialetti, R. Di Filippo, R. G. De Lorenzo, et al., "Optimizing nozzle travel time in proton therapy," in *2022 IEEE 35th International Symposium on Computer-Based Medical Systems (CBMS)*, IEEE, 2022, pp. 441–446.

10 G. Placidi, G. De Gasperis, F. Mignosi, M. Polsinelli, and M. Spezialetti, "Integration of a bci with a hand tracking system and a motorized robotic arm to improve decoding of brain signals related to hand and finger movements," in *Advances in Visual Computing: 16th International Symposium, ISVC 2021, Virtual Event, October 4-6, 2021, Proceedings, Part I*, Springer, 2021, pp. 305–315.

11 M. Spezialetti, F. Lapenna, P. Caianiello, et al., "Using deep learning for fast dose refinement in proton therapy," in *2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, IEEE, 2021, pp. 1783–1789.

12 F. A. D'Asaro, M. Spezialetti, L. Raggioli, and S. Rossi, "Towards an inductive logic programming approach for explaining black-box preference learning systems," in *Proceedings of the International Conference on Principles of Knowledge Representation and Reasoning*, vol. 17, 2020, pp. 855–859.

13 S. Sangiovanni, M. Spezialetti, F. A. D'Asaro, G. Maggi, and S. Rossi, "Administrating cognitive tests through hri: An application of an automatic scoring system through visual analysis," in *Social Robotics: 12th International Conference, ICSR 2020, Golden, CO, USA, November 14–18, 2020, Proceedings 12*, Springer, 2020, pp. 369–380.

14 G. R. Tizzano, M. Spezialetti, and S. Rossi, "A deep learning approach for mood recognition from wearable data," in *2020 IEEE international symposium on medical measurements and applications (MeMeA)*, IEEE, 2020, pp. 1–5.

15 P. Di Giamberardino, D. Iacoviello, G. Placidi, M. Polsinelli, and M. Spezialetti, "A brain computer interface by eeg signals from self-induced emotions," in *VipIMAGE 2017: Proceedings of the VI ECCOMAS Thematic Conference on Computational Vision and Medical Image Processing Porto, Portugal, October 18-20, 2017*, Springer, 2018, pp. 713–721.

16 G. Placidi, L. Cinque, F. Mignosi, M. Polsinelli, and M. Spezialetti, "Sparse sampling for magnetic resonance imaging," in *Proceedings on 15th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering*, 2018.

17 G. Placidi, L. Cinque, M. Polsinelli, and M. Spezialetti, "Characterization of a virtual glove for hand rehabilitation based on orthogonal leap controllers," in *Pattern Recognition Applications and Methods: 6th International Conference, ICPRAM 2017, Porto, Portugal, February 24–26, 2017, Revised Selected Papers 6*, Springer, 2018, pp. 190–203.

18 G. Placidi, L. Cinque, M. Polsinelli, and M. Spezialetti, "Forces calculation module for the leap-based virtual glove," in *Proceedings of the 2018 10th International Conference on Bioinformatics and Biomedical Technology*, 2018, pp. 67–72.

19 G. Placidi, M. Polsinelli, M. Spezialetti, and L. Cinque, "Bci driven by self-induced emotions: A multi-class study," in *2018 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, IEEE, 2018, pp. 1–6.

20 M. Polsinelli, P. A. Banchetti, A. Cacchio, et al., "Hand movement parameters calculated by the leap based virtual glove," in *2018 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, IEEE, 2018, pp. 1–6.

21 L. Cinque, A. De Santis, P. Di Giamberardino, et al., "Design of a classification strategy for light microscopy images of the human liver," in *Image Analysis and Processing-ICIAP 2017: 19th International Conference, Catania, Italy, September 11-15, 2017, Proceedings, Part I 19*, Springer, 2017, pp. 626–636.

22 G. Placidi, L. Cinque, P. Di Giamberardino, D. Iacoviello, and M. Spezialetti, "An affective bci driven by self-induced emotions for people with severe neurological disorders," in *New Trends in Image Analysis and Processing–ICIAP 2017: ICIAP International Workshops, WBICV, SSPandBE, 3AS, RGBD, NIVAR, IWBAAS, and MADiMa 2017, Catania, Italy, September 11-15, 2017, Revised Selected Papers 19*, Springer, 2017, pp. 155–162.

23 G. Placidi, L. Cinque, A. Petracca, M. Polsinelli, and M. Spezialetti, "A virtual glove system for the hand rehabilitation based on two orthogonal leap motion controllers.," in *ICPRAM 2017 – International Conference on Pattern Recognition Applications and Methods*, 2017, pp. 184–192.

24 G. Placidi, L. Cinque, A. Petracca, M. Polsinelli, and M. Spezialetti, "Iterative adaptive sparse sampling method for magnetic resonance imaging.," in *ICPRAM – International Conference on Pattern Recognition Applications and Methods*, 2017, pp. 510–518.

25 M. Spezialetti, D. Iacoviello, A. Petracca, and G. Placidi, "A virtual system for balance control assessment at home," in *ICTs for Improving Patients Rehabilitation Research Techniques: Third International Workshop, REHAB 2015, Lisbon, Portugal, October 1-2, 2015, Revised Selected Papers 3*, Springer, 2017, pp. 12–25.

26 G. Placidi, P. Di Giamberardino, A. Petracca, M. Spezialetti, and D. Iacoviello, "Classification of emotional signals from the deap dataset.," in *NEUROTECHNIX*, 2016, pp. 15–21.

27 D. Iacoviello, N. Pagnani, A. Petracca, M. Spezialetti, and G. Placidi, "A poll oriented classifier for affective brain computer interfaces.," in *NEUROTECHNIX*, 2015, pp. 41–48.

28 A. Petracca, M. Carrieri, D. Avola, et al., "A virtual ball task driven by forearm movements for neuro-rehabilitation," in *2015 International Conference on Virtual Rehabilitation (ICVR)*, IEEE, 2015, pp. 162–163.

29 G. Placidi, A. Petracca, M. Spezialetti, and D. Iacoviello, "Classification strategies for a single-trial binary brain computer interface based on remembering unpleasant odors," in *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, IEEE, 2015, pp. 7019–7022.

30 D. Avola, L. Cinque, S. Levialdi, A. Petracca, G. Placidi, and M. Spezialetti, "Innovative on-line handwriting identification algorithm based on stroke features," in *Computational Modeling of Objects Presented in Images. Fundamentals, Methods, and Applications: 4th International Conference, CompIMAGE 2014, Pittsburgh, PA, USA, September 3-5, 2014 4*, Springer, 2014, pp. 400–411.

31 D. Avola, L. Cinque, S. Levialdi, A. Petracca, G. Placidi, and M. Spezialetti, "Time-of-flight camera based virtual reality interaction for balance rehabilitation purposes," in *Computational Modeling of Objects Presented in Images. Fundamentals, Methods, and Applications: 4th International Conference, CompIMAGE 2014, Pittsburgh, PA, USA, September 3-5, 2014 4*, Springer, 2014, pp. 363–374.

32 G. Placidi, D. Avola, L. Cinque, G. Macchiarelli, A. Petracca, and M. Spezialetti, "Adaptive sampling and non linear reconstruction for cardiac magnetic resonance imaging," in *Computational Modeling of Objects Presented in Images. Fundamentals, Methods, and Applications: 4th International Conference, CompIMAGE 2014, Pittsburgh, PA, USA, September 3-5, 2014 4*, Springer, 2014, pp. 24–35.

33 S. Bisconti, M. Spezialetti, G. Placidi, and V. Quaresima, "Functional near-infrared frontal cortex imaging for virtual reality neuro-rehabilitation assessment.," in *CompIMAGE*, 2012, pp. 187–192.

34 M. Spezialetti, D. Avola, G. Placidi, and G. De Gasperis, "Movement analysis based on virtual reality and 3d depth sensing camera for whole body rehabilitation.," in *CompIMAGE*, 2012, pp. 367–372.

Speaker Activities

1 C. Epifanio, L. Forlizzi, F. Marzi, F. Mignosi, G. Placidi, and M. Spezialetti, *On the k-hamming and k-edit distances*, in *ICTCS 2023-Italian Conference on Theoretical Computer Science 2023 Proceedings of the 24th Italian Conference on Theoretical Computer*, CEUR, 2023, pp. 143–156.

2 M. Spezialetti, R. G. De Lorenzo, G. L. Gravina, et al., *Spread-out bragg peak in treatment planning system by mixed integer linear programming: A proof of concept*, in *2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS)*, IEEE, 2023, pp. 548–554.

3 M. Spezialetti, R. Di Filippo, R. G. De Lorenzo, et al., *Optimizing nozzle travel time in proton therapy*, in *2022 IEEE 35th International Symposium on Computer-Based Medical Systems (CBMS)*, IEEE, 2022, pp. 441–446.

4 M. Spezialetti, F. Lapenna, P. Caianiello, et al., *Using deep learning for fast dose refinement in proton therapy*, in *2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, IEEE, 2021, pp. 1783–1789.

5 G. R. Tizzano, M. Spezialetti, and S. Rossi, *A deep learning approach for mood recognition from wearable data*, in *2020 IEEE international symposium on medical measurements and applications (MeMeA)*, IEEE, 2020, pp. 1–5.

6 L. Cinque, A. De Santis, P. Di Giamberardino, et al., *Design of a classification strategy for light microscopy images of the human liver*, in *Image Analysis and Processing-ICIAP 2017: 19th International Conference, Catania, Italy, September 11-15, 2017, Proceedings, Part I 19*, Springer, 2017, pp. 626–636.

7 G. Placidi, L. Cinque, P. Di Giamberardino, D. Iacoviello, and M. Spezialetti, *An affective bci driven by self-induced emotions for people with severe neurological disorders*, in *New Trends in Image Analysis and Processing–ICIAP 2017: ICIAP International Workshops, WBICV, SS-*

PandBE, 3AS, RGBD, NIVAR, IWBAAS, and MADiMa 2017, Catania, Italy, September 11-15, 2017, Revised Selected Papers 19, Springer, 2017, pp. 155–162.

8 G. Placidi, L. Cinque, A. Petracca, M. Polsinelli, and M. Spezialetti, *A virtual glove system for the hand rehabilitation based on two orthogonal leap motion controllers.* In *ICPRAM 2017 – International Conference on Pattern Recognition Applications and Methods*, 2017, pp. 184–192.

9 G. Placidi, L. Cinque, A. Petracca, M. Polsinelli, and M. Spezialetti, *Iterative adaptive sparse sampling method for magnetic resonance imaging.* In *ICPRAM – International Conference on Pattern Recognition Applications and Methods*, 2017, pp. 510–518.

10 M. Spezialetti, D. Iacoviello, A. Petracca, and G. Placidi, *A virtual system for balance control assessment at home*, in *ICTs for Improving Patients Rehabilitation Research Techniques: Third International Workshop, REHAB 2015, Lisbon, Portugal, October 1-2, 2015, Revised Selected Papers 3*, Springer, 2017, pp. 12–25.

Participation to Research Groups

- 2020 – present • Artificial Intelligence & Information Mining (AIIM) Group (https://aiimlab.org/).
- 2020 – present • Research group on optimization of computational resources in TPS (Treatment Planning System) for hadron therapy, coordinated by Prof. Filippo Mignosi.
- 2022 – present • Participation in the activities of the European research project ECSEL-JU 2020 RIA AIDOaRt (AI-augmented automation for efficient DevOps, a model-based framework for continuous development At RunTime in cyber-physical systems), funded by Electronic Component Systems for European Leadership Joint Undertaking (ECSEL-JU) under agreement No 101007350.
- 2020 – 2025 • Research group on algorithms, data structures, operational research, artificial intelligence of the Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila, coordinated by Prof. Claudio Arbib.
- 2019 – 2020 • Research group coordinated by Prof. Silvia Rossi within the project "Automatic evaluation of the experience during multi-agent human-machine and human-robot interaction, using biometric signals for process features extraction".
- 2018 – 2019 • Research group coordinated by Prof. Viviana Betti within the project ERC Starting Grant HANDmade (How natural hand usage shapes behavior and intrinsic and task-evoked brain activity) under agreement No 759651.
- 2012 – 2018 • Research group in biomedical informatics "A²VILab", coordinated by Prof. Giuseppe Placidi.

Other Activities

- 2026 • **Organizer:** Web&graph2026, Workshop on Web & Graphs, Responsible Intelligence, and Social Media (WSDM26), co-located with 19th ACM International Conference on Web Search and Data Mining (WSDM26)
- 2026 • **Session Chair:** 19th ACM International Conference on Web Search and Data Mining (WSDM26)
- 2023 – present • **Associate Editor:** Frontiers in Neurology: Artificial Intelligence in Neurology
- 2023 • **Organising Committee Member, Session Chair, Local Chair:** IEEE 36th International Symposium on Computer Based Medical Systems (CBMS2023)
- 2023 – 2024 • **PC Member:** IEEE 36/37th International Symposium on Computer Based Medical Systems (CBMS2023-2024)
- 2023 – present • **PC Member:** 18th/22nd International Conference on Computer Vision Theory and Applications (VISAPP2022-2027)
- 2023 – 2025 • **PC Member:** 26/28th European Conference on Artificial Intelligence (ECAI23-25)

Other Activities (continued)

- 2022 • **PC Member:** Joint Workshop (BRIO Workshop (Bias, Risk and Opacity in AI), the ME&E-LP Workshop (Machine Ethics & Explainability - the Role of Logic Programming), and the AWARE AI Workshop (Ethics and AI, a two-way relationship)) in conjunction with AIxIA 2022. (BEWARE2022)
- 2021 • **PC Member:** 1st Workshop on Machine Ethics and Explainability - The Role of Logic Programming (MEandE-LP2021)
- 2018 – present • **Reviewer for International Journals:** Image and Vision Computing, IEEE Transactions on Cognitive and Developmental Systems, Biomedical Signal Processing and Control, Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization, Computers in Biology and Medicine, Computer Methods and Programs in Biomedicine Update, Brain Sciences, Frontiers in Robotics and AI, Frontiers in Computational Neuroscience, Data in Brief.

Research Projects

- 2025 • **Project:** Definition of rehabilitation protocols for masticatory function, and of sEMG data acquisition and analysis. Feasibility study for the home use of wearable devices
  Type: Joint project DISIM-MESVA, University of L'Aquila
  Amount: 8000 €
- 2024 • **Project:** Compression, Entropy and Deep Learning
  Type: Research Project, University of L'Aquila
  Amount: 12000 €
- 2021 • **Project:** Design and development of an optimization pipeline for proton therapy
  Type: Research Starting Project, University of L'Aquila
  Amount: 3000 €
- 2018 • **Project:** From the brain to the hand. Analysis and classification of movement encoding and functional synergies at the neuronal level and in spontaneous activity
  Type: Research Starting Project, University "La Sapienza" di Roma
  Amount: 2000 €